

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

## > Data

↳ 6 cells hidden

## ✓ Test case

### ✓ 1

```
v=4.3          # m/s
m_dot_air=v*rho_air*A_air # kg/s
q=m_dot_water*cp_water*(T_in_water-T_out_water)
T_out_air=T_in_air+q/(m_dot_air*cp_air)
```

q

6.976666666666667

T\_out\_air

28.506747771695807

T\_out\_air-T\_in\_air

3.5067477716958066

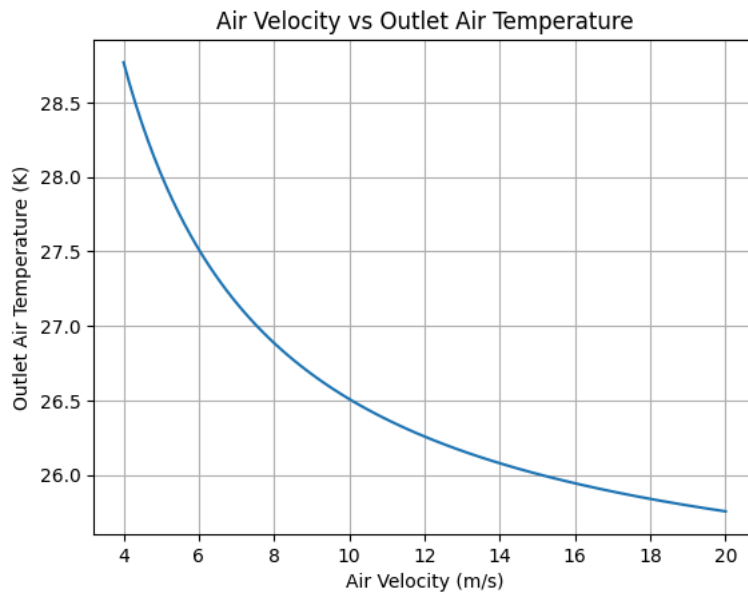
### ✓ 2

Double-click (or enter) to edit

```
v_arr=np.linspace(4,20,100)
```

```
m_dot_air_arr=v_arr*rho_air*A_air
t_out=T_in_air+q/(m_dot_air_arr*cp_air)
```

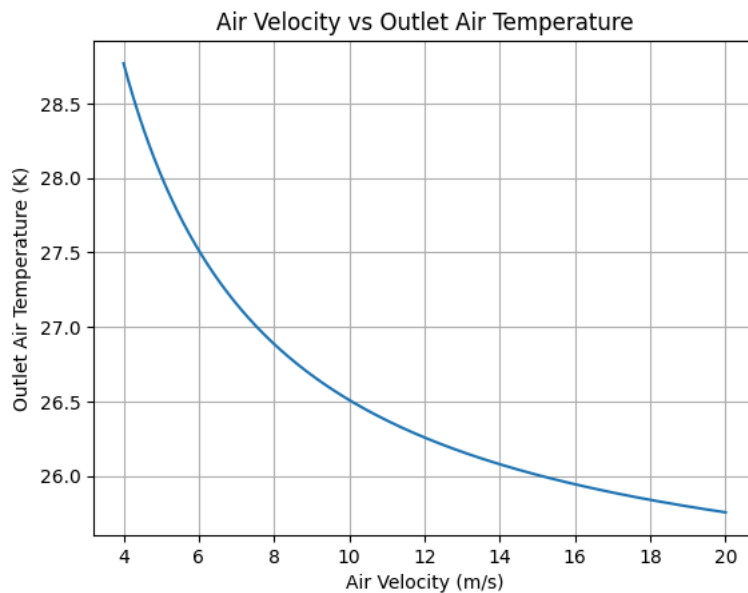
```
plt.plot(v_arr, t_out)
plt.xlabel('Air Velocity (m/s)')
plt.ylabel('Outlet Air Temperature (K)')
plt.title('Air Velocity vs Outlet Air Temperature')
plt.grid()
plt.show()
```



3

```
v_arr=np.linspace(4,20,100000)
m_dot_air_arr=v_arr*rho_air*A_air
t_out=T_in_air+q/(m_dot_air_arr*cp_air)
t_out=np.array(t_out)
```

```
plt.plot(v_arr, t_out)
plt.xlabel('Air Velocity (m/s)')
plt.ylabel('Outlet Air Temperature (K)')
plt.title('Air Velocity vs Outlet Air Temperature')
plt.grid()
plt.show()
```



black radiator

```
r_l=0.242
r_w=0.12
r_h=0.024
t_l=r_l
t_w=r_w
f_l=r_w
f_w=0.0015
f_h=0.009
f_n=95*2*13
```

```
t_n=12
t_h=0.0009
```

```
coolant_area,air_area=sa(r_l, r_w, r_h, t_l, t_w, t_h, t_n, f_l, f_w, f_h,f_n)
air_area_1=f_h*f_w*f_n
vel=10
T_in=40
m=vel*air_area_1*rho_air
Q=abs(vel*air_area*rho_air*cp_air*(8))
Q_w=m_dot_water*cp_water*(T_in_water-T_out_water)
```

m

0.38726883

air\_area

6.224399999999999

$T=Q_w/(m)$   
T

18.015048271937264

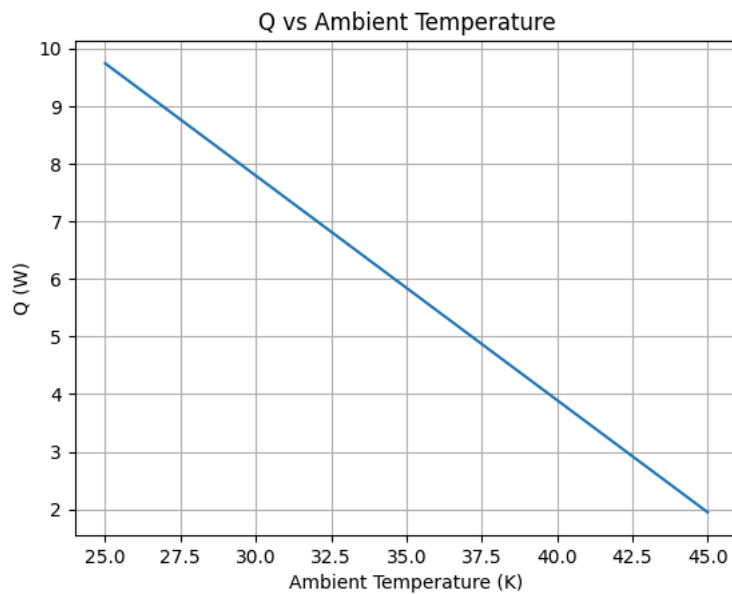
Q

581.7913815168

## Q vs ambient Temperature

T\_air\_out=50

```
T_amb=np.linspace(25,45,100)
Q_dot=rho_air*cp_air*(T_air_out-T_amb)*air_area_1*vel
plt.plot(T_amb, Q_dot)
plt.xlabel('Ambient Temperature (K)')
plt.ylabel('Q (W)')
plt.title('Q vs Ambient Temperature')
plt.grid()
plt.show()
```



```
T_rad_in=70
T_rad_out=60
T_air_in=25
Q_dot=m_dot_water*cp_water*(T_rad_in-T_rad_out)
T_air_out=T_air_in+Q_dot/(m*cp_air)
```

```
theta_1 = T_rad_in - T_air_out
theta_2 = T_rad_out - T_air_in
```

```
theta_1, theta_2
```

```
(27.09239734399874, 35)
```

```
LMTD=(theta_1-theta_2)/np.log(theta_1/theta_2)
```

```
LMTD
```

```
np.float64(30.877624781578774)
```

```
U=Q_dot/(coolant_area*LMTD)  
U
```

```
np.float64(0.3217741706492126)
```

```
T=50
```

```
T_amb=np.linspace(25,45,50)  
theta_1=T_rad_in-T  
theta_2=T_rad_out-T_amb  
LMTD=(theta_1-theta_2)/np.log(theta_1/theta_2)  
Q_out=U*coolant_area*LMTD  
plt.plot(T_amb, Q_out)  
plt.xlabel('Ambient Temperature (K)')  
plt.ylabel('Q (kW)')  
plt.title('Q vs Ambient Temperature')  
plt.grid()  
plt.show()
```

